

Algebra II with Trigonometry

Chapter 5.4

Study guide

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Study Guide: Inverse Trigonometric Functions and Right Triangles

1. Short Summary

This document introduces the inverse trigonometric functions—arcsine, arccosine, and arctangent—exploring their definitions, domains, and ranges. It details how to use these functions to solve for unknown angles in right triangles and to evaluate various trigonometric expressions. The document includes both exact and calculator-based evaluation of inverse trig functions, applications to word problems, and rewriting expressions algebraically.

2. Core Definitions, Theorems, and Formulas

Definitions & Notation:

- **Inverse Sine (arcsin):** $y = \sin^{-1}(x) \iff \sin(y) = x$
- **Domain:** $[-1, 1]$
- **Range:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- **Inverse Cosine (arccos):** $y = \cos^{-1}(x) \iff \cos(y) = x$
- **Domain:** $[-1, 1]$
- **Range:** $[0, \pi]$
- **Inverse Tangent (arctan):** $y = \tan^{-1}(x) \iff \tan(y) = x$
- **Domain:** $\mathbb{R}$
- **Range:** $(-\frac{\pi}{2}, \frac{\pi}{2})$

- **Other inverses:**

- $\sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right), x \geq 1$
- $\csc^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right), x \geq 1$
- $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right), x > 0$

Common Right Triangle Relationships:

- $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
- $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

Solving for Angles:

- If two sides are known, angle θ is:
 - $\theta = \sin^{-1}\left(\frac{y}{r}\right)$
 - $\theta = \cos^{-1}\left(\frac{x}{r}\right)$
 - $\theta = \tan^{-1}\left(\frac{y}{x}\right)$
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3. Step-by-Step Study Guide

A. Understanding Inverse Trigonometric Functions

Why restricted domains? The basic trigonometric functions are not one-to-one unless restricted, so their inverses must have specific domains and ranges:

- **arcsin:** only gives angles between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.
- **arccos:** only gives angles between 0 and π .
- **arctan:** gives angles between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ (not including endpoints).

B. Using Inverses to Find Angles in Right Triangles

Given two sides of a right triangle: 1. Calculate ratio relevant to the function (e.g., for $\sin$ use opposite/hypotenuse). 2. Apply the appropriate inverse trig function, e.g.:

- $\theta = \sin^{-1}\left(\frac{3}{5}\right)$
- Use calculator for decimal/radian/degree value.

C. Evaluating Inverse Trig Expressions

- **Exact values:** For common ratios (like 0, 1, $-\frac{1}{2}$), use knowledge of the unit circle.
- **Calculator values:** For arbitrary values, use a calculator and round as instructed.

Example:

- $\sin^{-1}(0) = 0$
- $\cos^{-1}(-1) = \pi$
- $\tan^{-1}(1) = \frac{\pi}{4}$

D. Rewriting and Composing Functions

- Start with basic relationships, possibly sketch a triangle.
- For expressions like $\cos(\sin^{-1} x)$, draw a triangle where $\sin \theta = x$, use the Pythagorean Theorem to find other sides, and solve for $\cos \theta$.

E. Applications: Word Problems

- **Angle of Elevation:** Use tangent: $\tan \theta = \frac{\text{height}}{\text{base}} \Rightarrow \theta = \tan^{-1} \left(\frac{\text{height}}{\text{base}} \right)$.
- **Expressing variables:** If you have the angle, $h = (\text{base}) \tan \theta$. If you have the height, $\theta = \tan^{-1} \left(\frac{h}{\text{base}} \right)$.

4. Practice Problems with Answers

1. Find the exact value: $\sin^{-1} \left(\frac{1}{2} \right)$ **Answer:** $\frac{\pi}{6}$
2. Use a calculator to approximate: $\tan^{-1}(2)$ (in degrees, to 1 decimal place) **Answer:** 63.4°
3. Find $\cos(\sin^{-1}(\frac{3}{5}))$ exactly. **Solution:** Let $\theta = \sin^{-1}(\frac{3}{5}) = \sin \theta = \frac{3}{5}$
Draw a triangle: opposite=3, hypotenuse=5, adjacent= $\sqrt{5^2 - 3^2} = 4$.
 $\cos \theta = \frac{4}{5}$ **Answer:** $\frac{4}{5}$
4. Given a right triangle with sides 7 (adjacent to θ), 24 (opposite), find θ in radians. **Solution:** $\theta = \tan^{-1} \left(\frac{24}{7} \right) \approx 1.287$ radians.

5. A ladder leans against a wall, reaching a height of 10 ft when its base is 6 ft from the wall. Find the angle the ladder makes with the ground (to 1 decimal in degrees). $\theta = \tan^{-1} \left(\frac{10}{6} \right) \approx 59.0^\circ$

5. Assumptions & Ambiguities Noted

- The symbol “R” for the domain of $\tan^{-1}$ is interpreted as $\mathbb{R}$, the set of all real numbers.
 - Angles are expressed in both degrees and radians, as per examples; unless specified, radians are the default.
 - Roots and values (like $\sqrt{3}/2$, etc.) are recognized from unit circle knowledge.
 - For some problems requiring calculator answers, "approximately" means round as directed (to three decimal places or one decimal place).
 - For composition problems ($\cos(\sin^{-1} x)$), triangle constructions are implied.
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